

Evaluating performances of various CNN architectures for multi-class classification of rotten fruits

Krishiv Dakwala

Department of Information Technology
Pune Institute of Computer Technology
Pune, India
krishivdakwala@gmail.com

Vishwajit Shelke

Department of Information Technology
Pune Institute of Computer Technology
Pune, India
vishwajitds@gmail.com

Pranav Bhagwat

Department of Information Technology
Pune Institute of Computer Technology
Pune, India
pranav221b@gmail.com

Dr. Anant M. Bagade

Department of Information Technology
Pune Institute of Computer Technology
Pune, India
ambagade@pict.edu

Abstract— In recent years, Artificial Intelligence (AI) has been at the crux of many advancements across different industries such as E-Commerce, Finance, and Healthcare. The introduction of new Machine Learning techniques in the agri-food industry could become a turning point. The quality of the produce is one of the major factors that drives this industry, and it plays an important role in consumer consumption thereby affecting the sale of the produce. For a country like India, which heavily depends on agriculture, the introduction of modern methods in agriculture is vital for increasing efficiency and thus increasing the amount and quality of produce. AI-driven autonomous systems could cut down the manual labor which will hence reduce processing times and can have a huge impact on processing expenditure as well as sales. Deep Learning along with Computer Vision can be used to analyze the quality of fruits and vegetables. It can also be used to grade the goods according to their freshness and can even identify rotting fruits and vegetables. Traditionally, manual labor is used to perform the above-mentioned tasks but with the introduction of the proposed technology, we can cut down the costs and reduce the processing time. We plan to implement various pre-trained CNN architectures using Transfer Learning to implement multi-class classification of rotten fruits. Finally, we use performance speed and accuracy as the main metrics to compare the models and showcase their usefulness in this domain.

Keywords – Computer Vision, Deep Learning, Convolutional Neural Networks, Artificial Intelligence, Image Classification

I. INTRODUCTION

Fruit spoiling has a huge economic impact; it is estimated that roughly a third of the fruits sold are rotten and lead to wastage of food and poor sales. The sale of fruits is impacted as customers refuse to buy spoiled fruits. The reduced concentrations of amino acids, vitamins, sugar/glucose, and other nutrients inevitably raise public concerns about edibility issues, prompting discussions on how to prevent or slow the decaying process.

E-commerce has exploded in popularity over the last few years. People are increasingly purchasing goods online,

including their daily requirement of fruits and vegetables. The problem at hand is that the consumer is unaware of the quality of the fruits before purchasing them. Consumers may receive rotten or old fruits, resulting in the wastage of food and money. Consumers will receive the highest quality items if automated technologies can replace manual segregation and removal of rotting food. This can boost sales and reduce wastage. Segregation into fresh and rotten right after harvesting the fruits and vegetables might be a better way to handle this problem [1]. This could cut total waste as well as the time and effort required to manually separate rotten produce. This can save expenses, improve quality, and raise revenues overall.

Rot detection in fruits is significant because of the importance of food in people's lives and contribution to the economy, yet the manual operation for the same is time consuming and inefficient. Automation of grading by computerized methods is thought to be the answer to this problem.

When fruit deteriorates, established research evidence demonstrates that it goes through a sequence of metabolic transformations that result in changes in its physical conditions, such as color and shape. The majority of these characteristics can be recorded. The most cost-effective solution is projected to be a computer vision-based technique. In this paper, we propose an efficient and effective computer vision system based on cutting-edge deep learning techniques, as well as a cost-effective solution for automating the rot detection process in fruits

The aim of this paper is to study various Convolutional Neural Network (CNN) architectures [2] and identify the best multi-class classification model which is needed for automating the process of classifying fruits as rotten or not. To discover the best model for detecting rotten produce, we trained, tested, and compared the performance of several deep learning models, including ResNet [3], VggNet [4], DenseNet [5], Xception [6], and NASNet [7].

II. LITERATURE REVIEW

This section elaborates on the existing methodologies to detect the infected leaves. Nazrul Ismail [8] reviews methodologies that perform automatic visual inspection of fruits' freshness and appearance. He compared various deep learning models with respect to accuracy but the speed of the model is not taken into account and some models such as VggNet and Xception are not considered.

Yuhang Fu [9] reviews various architectures such as AlexNet, VggNet, ResNet, Inception, Google Net, etc and proposed a grading system for fruit and vegetable freshness. The dataset used here was small considering the number of classifications made and some of the newer versions of the architectures (such as Vgg19, ResNet50, ResNet v2 etc.) were not reviewed. Xception was not considered for this review.

Richa Shah [10] proposed a clustering-based framework which aims to predict the quality of fruits and vegetables. One of the major drawbacks of the proposed model is the use of ANN instead of CNN which reduces the performance speed greatly.

Mrs. Pushpa B R [11] proposed and evaluated an image processing-based approach to detect fruit disease where features are directly extracted using image processing which leaves no room for deep learning models.

Jose Naranjo-Torres [12] reviewed various CNN architectures – AlexNet, Vgg16, MobileNet [13], Inception V3 [14], ResNet 50. A single fruit dataset for quality classification (Apple-NDDA) was used here. Some of the newer versions of the architectures (Vgg19, ResNet v2) were not taken into consideration. Xception architecture was not taken for review.

III. PROPOSED METHODOLOGY

The process diagram shown in Figure 1. shows the methodology followed to achieve the desired task.

The dataset selected for the training purpose can be found on Kaggle, named 'Fruits fresh and rotten for classification' [15]. The dataset contains images of fresh and rotten apple, bananas, and oranges, thus there are 6 classes in total. There are 5501 training images and 1384 test images within the dataset.

The dataset selected is already cleaned and ready for further processing. Before further processing we randomly flip the images vertically and horizontally in order to prevent overfitting. All the images are then rescaled. In the next step of data augmentation, we use transfer learning. Transfer learning is reusing a pretrained model on a new problem, with the model leveraging knowledge from a previous task to increase generalization on a subsequent job.

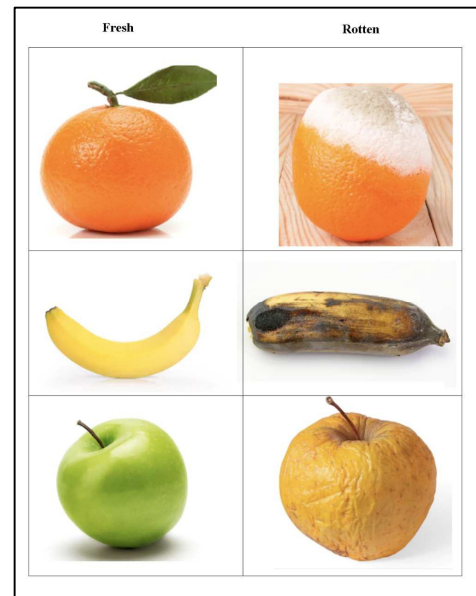


Figure 1. Sample images from the dataset.

We will be using the following architectures: VggNet16, VggNet19, ResNet, ResNet50, ResNet v2, NasNet, Xception and DenseNet. Next, we must rescale our image to 1/255, which normalizes and transforms each pixel in the range of 0 to 255 to the range of 0 to 1. Since different images have different pixel ranges, we do this to treat all of them the same way. Now, a horizontal and vertical flip will be applied to our train batches, but not to our testing data. We don't want to supplement our testing data; instead, we want to make sure it's completely new. Finally, we will reshape the data and the data is ready for further processing.

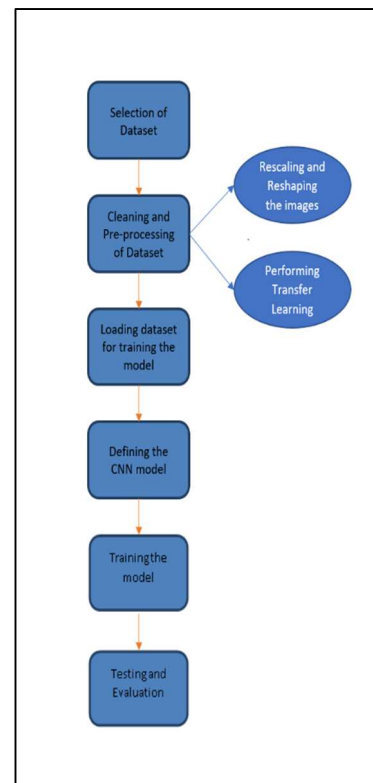


Figure 2: Process Diagram

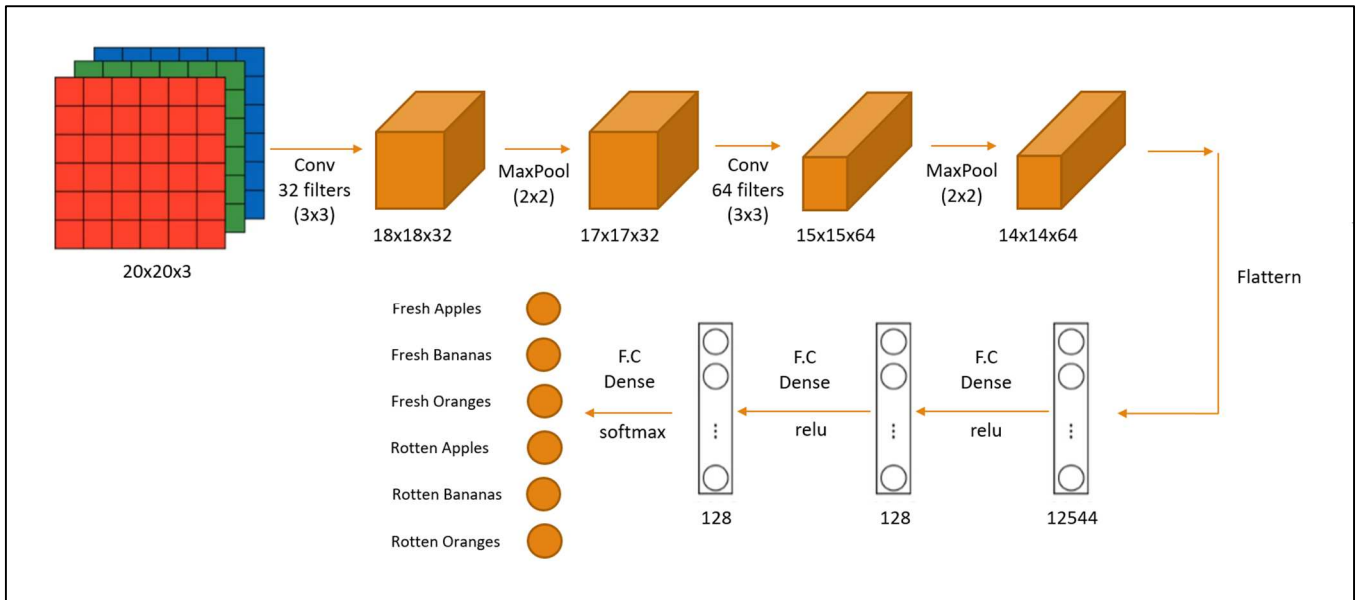


Figure 3: Model Architecture

For constructing the convolutional neural network, we will add the following layers: We will add a convolutional layer as our first layer. It will have 32 filters with a filter size of (3, 3). We will use Rectified Linear Unit (ReLU) as the activation function. After this we will add a max pooling layer with a pool size of (2, 2). Now we will need to add another Conv2D layer but this time having 64 filters with a size of (3,3), continuing with the ReLU activation. We will also add another MaxPooling2D layer that will have a pool size of (2, 2).

Next, we will create our dense layers after using the Flatten function to flatten our input. We will add two dense layers with 128 neurons each. For these layers, we will continue to use the ReLU activation function. The final layer will contain 6 neurons for 6 classes (fresh apples, fresh bananas, fresh oranges, rotten apples, rotten bananas, rotten oranges). This layer uses softmax activation function.

For compiling and training our model we will use 'Adam' as our optimizer and 'categorical cross entropy' for our loss function. Since we are dealing with a multi-class classification problem, we chose this as our loss function. Finally, after the model is trained, we will test the model on the test batches to calculate model metrics such as speed and accuracy.

Note that we will use these 6 steps to iterate over all the chosen architectures (Vgg16, Vgg19, ResNet, ResNet50, ResNet v2, NasNet, Xception and DenseNet). The only change would be during the second step where we apply transfer learning on the dataset.

IV. RESULTS

For the results, please refer to Table 1. The main metrics chosen to compare the 8 CNN architectures are: Performance Time (in milliseconds) - the amount of time taken by the model to classify the test dataset and Accuracy (in percentage). It can be inferred from the table that ResNet50 has the best performance with the best combination high accuracy and low performance time. VggNet16, ResNet and VggNet19 trail closely behind with

similar accuracies but more variation in their performance times.

Sr.	Architecture	Performance Time (ms)	Accuracy (in %)
1	VggNet16	5699	98.41
2	ResNet	4726	98.12
3	NasNet	5143	93.79
4	Xception	3753	92.34
5	ResNet50	3750	98.27
6	ResNet v2	4031	95.38
7	DenseNet	3892	96.53
8	VggNet19	5421	97.69

Table 1. Results obtained after applying Transfer Learning on the 8 CNN architectures chosen.

The graphical representation for the above table:

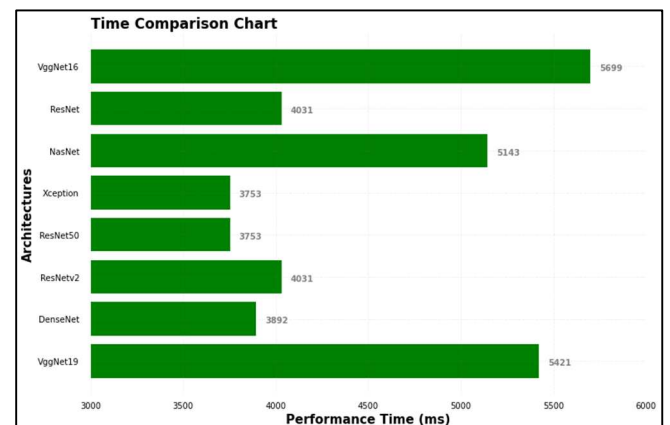


Figure 4: Performance Time Comparison Chart

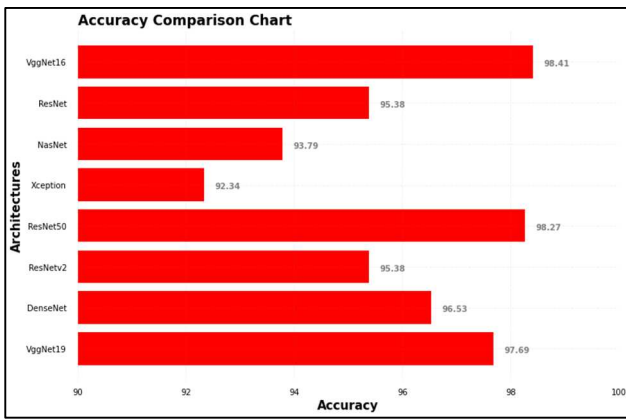


Figure 5: Accuracy Comparison Chart

V. CONCLUSION AND FUTURE SCOPE

The goal of this paper is to evaluate various CNN (Convolutional Neural Network) architectures when used to classify different fruits as rotten or fresh. We looked at different architectures such as ResNet, VggNet, DenseNet, Xception, and NasNet. The evaluation results revealed the fastest model to be ResNet50 and the most accurate as VggNet16. The transition from human labour to AI-based systems can lower costs, reduce processing time, boost efficiency, and hence boost overall profits. The applications of this technology can be anywhere within the supply chain from the farmer to the consumer. It can help in the early identification of rot in fruits which can help manage the supply and transport of these goods within the supply chain. It will also reduce food wastage at the consumer's end as the farmer can compost the rotten fruits as fertilizer. In an era of population inflation, proper food management at every step of the supply chain is paramount for the well-being of people and this project can be scaled to work in real-time and automate the process of rot detection in fruits and other produce. In this paper, we compared and tested several architectures that could be used to implement the suggested model, determining their speeds and accuracies.

VI. REFERENCES

- [1] Onur, A. Y. A. N., et al. "The detection of spoiled fruits on a conveyor belt using image processing techniques and OPC server software." *International Journal of Computational and Experimental Science and Engineering* 4.1 (2018): 11-15.
- [2] O'Shea, Keiron, and Ryan Nash. "An introduction to convolutional neural networks." *arXiv preprint arXiv:1511.08458* (2015).
- [3] He, Kaiming, et al. "Deep residual learning for image recognition." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.
- [4] Simonyan, Karen, and Andrew Zisserman. "Very deep convolutional networks for large-scale image recognition." *arXiv preprint arXiv:1409.1556* (2014).
- [5] Huang, Gao, et al. "Densely connected convolutional networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.
- [6] Chollet, François. "Xception: Deep learning with depthwise separable convolutions." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.

- [7] Zoph, Barret, et al. "Learning transferable architectures for scalable image recognition." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.
- [8] Ismail, Nazrul, and Owais A. Malik. "Real-time visual inspection system for grading fruits using computer vision and deep learning techniques." *Information Processing in Agriculture* 9.1 (2022): 24-37.
- [9] Fu, Yuhang. *Fruit freshness grading using deep learning*. Diss. Auckland University of Technology, 2020.
- [10] Pushpa, B. R., T. K. Meghana, and K. H. Tripulla. "Detection and classification of fungal disease in fruits using image processing technique." *International journal of applied engineering research (IJAER)* 10.55 (2015).
- [11] Shah, Richa P. et al. "Quality Analysis of Fruits and Vegetables using Machine Learning Techniques." (2018).
- [12] Re W. C. Seng and S. H. Mirisae, "AA New Method for Fruits Recognition System", 2009 International Conference on Electrical Engineering and Informatics, 5-7 August 2009, Selengore, Malaysia.
- [13] Howard, Andrew G., et al. "Mobilenets: Efficient convolutional neural networks for mobile vision applications." *arXiv preprint arXiv:1704.04861* (2017).
- [14] Szegedy, Christian, et al. "Going deeper with convolutions." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2015.
- [15] <https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification>